ROADMAP

Your 46-Day Path from Absolute Beginner to Edge AI Engineer

Raspberry Pi 4 (4 GB) + Windows laptops + two Pis + 2 hours a day (about 9 weeks at 5 days a week)

The plan in one picture

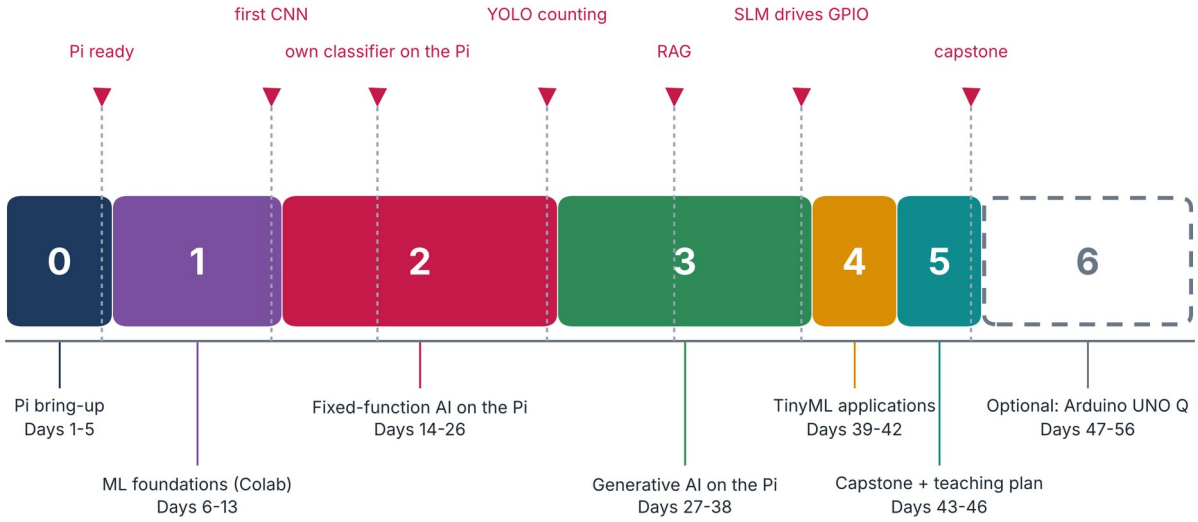


Figure 1. Six phases, 46 core days, and an optional Arduino UNO Q track. Red flags are the milestones you will be able to demonstrate.

Phase	Days	What you learn	What you need
0 Pi bring-up	1-5	Flash, boot, SSH, Linux basics, camera, Python + Jupyter on the Pi	Pi 4, microSD, power, laptop
1 ML foundations (Colab)	6-13	Neurons, DNNs, CNNs, datasets, metrics, overfitting, Edge Impulse intro	Laptop + Google Colab, no Pi needed
2 Fixed-function AI on the Pi	14-26	Image classification, object detection (SSD, FOMO), YOLO, counting objects	Camera (Pi camera or USB webcam), Edge Impulse account
3 Generative AI on the Pi	27-38	RNN text, distillation, small language models (Ollama), RAG, agents, vision-language, audio, GPIO, IoT control	USB mic + speaker, breadboard, LEDs, button, sensors
4 TinyML applications	39-42	Anomaly detection, keyword spotting, image apps on microcontrollers, responsible AI	Concepts only (no microcontroller board needed)
5 Capstone + teaching plan	43-46	Your own project, write-up, and the plan to teach it to students	Whatever the project needs
6 Optional: Arduino UNO Q	47-56	The same ideas on a different board (after you master the Pi and ML)	Arduino UNO Q (not owned yet)

How to use the plan

- **One day = about 2 hours.** Read the tutorial, watch the animation, do the lab, then print the sketch notes and add your own comments.

- **Two Pis, two laptops: do every day twice.** Pass 1 is the learning pass on one Pi (about 2 hours). Pass 2 is the consolidation pass on the other Pi, from your notes only, which should take less time. Alternate which Pi you learn on. This catches mistakes that hide behind copy-paste.
- **Name your Pis differently** (for example `raspi-office` and `raspi-home`) and keep the same user name on both. Keep code and notes in the GitHub repo and use `git pull`, so both Pis and both laptops stay in sync.
- **Phase 1 (ML foundations) needs no Pi**, only Google Colab, so it is a good block to do when you are travelling.
- **Book page numbers** below are PDF pages of the book file (printed page + 1). The splits of long chapters across days are estimates; I will adjust them as we go.

CAREFUL Honest limits of this plan

- **Pi 4 with 4 GB:** the vision chapters will run fine. The language-model chapters (Phase 3) were written for a Pi 5 and will be slower here, and only small models fit. We will choose model sizes to match your board and I will report slower-than-book results as they are.
- **Software versions:** a newer Raspberry Pi OS may ship a newer Python than the book used, and TensorFlow Lite can be picky about that. Day 4-5 includes the check.
- **Online services** (Colab, Edge Impulse, model downloads) need Internet access, and your office network may restrict some of them. Plan to use a phone hotspot on those days if needed.
- **UNO Q chapters** are from a different online book and course slides that I have not read yet; the RAG classes for it are marked 'to be developed' in the course README.

What to prepare, and when

By day	Item	Why
1	Pi 4, microSD (32 GB+), USB-C power 5 V / 3 A, card reader, Wi-Fi details	Everything in Phase 0
4	Pi camera module (CSI) **or** a USB webcam	Camera test and all vision projects
6	Google account (Colab)	ML foundations
13	Edge Impulse account	Training in Phase 2 (check the current free tier)
27	Internet on the Pi for model downloads; enough free space on the microSD	Ollama and models
35	USB microphone and speaker	Audio pipeline
36	Breadboard, jumper wires, LEDs + resistors, push button, DHT temperature sensor, BMP280 pressure sensor	GPIO and sensors (the repo scripts use these; confirm the exact parts in the chapter before buying)
47+	Arduino UNO Q (optional)	Alternative hardware track

Day-by-day plan

C = class number in the course repositories (IESTI05 = Edge AI, IESTI01 = TinyML). **Book pp.** = PDF pages of 'Edge-AI-Engineering book.pdf'. A dash means the day is based on slides, notebooks and docs rather than the book.

Day	Topic	Source	Book pp.	You will have
Phase 0 - Pi bring-up (days 1-5)				
1	Meet the Pi 4, flash the microSD, first SSH	IESTI05 C3	20-21, 25-27, 29-39	Pi answers over SSH
2	Remote access: file transfer, remote desktop, two Pis and two networks	IESTI05 C3	38-44, 56-61	Copy files both ways; desktop opened remotely
3	Linux command line	IESTI05 C2 (slides + linux_intro.pdf)	-	Navigate, edit, install, htop
4	Camera, swap memory, updates	IESTI05 C3 slide 10	43-55, 62-64	First photo taken on the Pi
5	Python virtual environment + Jupyter on the Pi	IESTI05 C4	65-75	Notebook running on the Pi
Phase 1 - ML foundations (Colab) (days 6-13)				
6	Colab and Python review; what is TinyML	IESTI01 C2-C4b	-	Colab notebook running
7	The machine-learning paradigm	IESTI01 C5	-	First model fitted
8	Neural-network building blocks	IESTI01 C6	-	A small network in TensorFlow
9	Regression with a DNN	IESTI01 C7	-	Regression model + plot
10	Classification with a DNN	IESTI01 C8	-	Classifier + confusion matrix
11	Datasets, splits and metrics	IESTI01 C9	-	Train/validation/test done right
12	Convolutions; image classification with a CNN	IESTI01 C10-C11	-	CNN trained in Colab
13	Edge Impulse intro (Cifar-10); preventing overfitting	IESTI01 C12-C13	-	Model trained in Edge Impulse
Phase 2 - Fixed-function AI on the Pi (days 14-26)				
14	TensorFlow Lite + MobileNetV2 inference	IESTI05 C5	75-86	Classify an image on the Pi
15	Live camera classification; a model trained from zero	IESTI05 C5	86-93	Live classification from the camera
16	Custom classifier: goal and data collection	IESTI05 C6	94-102	Your own dataset
17	Train in Edge Impulse	IESTI05 C7	103-111	Trained, quantised model
18	Test, deploy and run live on the Pi	IESTI05 C7	112-130	First custom classifier on the Pi
19	Object detection fundamentals	IESTI05 C8	131-145	Pre-trained detector on an image
20	Pre-trained detectors and live stream	IESTI05 C8	146-153	Live detection
21	Custom detection: data and labeling	IESTI05 C9	154-168	Labeled dataset
22	SSD MobileNet in Edge Impulse; deploy	IESTI05 C10	169-187	Detector on the Pi
23	FOMO (lightweight detection)	IESTI05 C10	188-202	FOMO model on the Pi
24	YOLO with Ultralytics on the Pi	IESTI05 C11	203-219	YOLO inference, NCNN export
25	YOLO: custom training and deployment	IESTI05 C12	220-237	Custom YOLO model
26	Counting objects (bees) + Part 1 mini-project	IESTI05 C13-C14, Part 1 assignment	238-264	Counting pipeline
Phase 3 - Generative AI on the Pi (days 27-38)				
27	Text generation with an RNN (VerneBot)	IESTI05 C15	265-289	Tiny text generator
28	Knowledge distillation in practice	book chapter	290-308	Distilled model
29	Small language models: first model with Ollama	IESTI05 C16-C17	309-335	SLM answering on the Pi
30	Ollama with Python	IESTI05 C18	336-360	Python chat script
31	Function (tool) calling	IESTI05 C19	361-375	SLM calling a Python function
32	Retrieval-augmented generation (RAG)	IESTI05 C20	376-388	Question answering over your documents

Day	Topic	Source	Book pp.	You will have
33	Agents with small models	IESTI05 C20a	560 onward	Simple tool-using agent
34	Vision-language models (Florence-2)	book chapter	389-433	Image captioning on the Pi
35	Audio and vision AI pipeline	book chapter	434-458	Speech in, answer out
36	GPIO and physical computing: LEDs, buttons	IESTI05 C21	459-485	LED and button under code control
37	Sensors and notebooks	IESTI05 C21	486-510	Sensor readings logged
38	SLMs for IoT control	IESTI05 C22	511-557	A language model switching hardware
Phase 4 - TinyML applications (days 39-42)				
39	Anomaly detection and K-means	IESTI01 C21-C22	-	Anomaly detector (concept + notebook)
40	Keyword spotting (concepts)	IESTI01 C23-C24	-	KWS pipeline understood
41	Image apps on microcontrollers; data collection	IESTI01 C25-C27	-	Person detection concept
42	Responsible AI; going further with Edge AI	IESTI01 C28-C29	-	Checklist for responsible deployment
Phase 5 - Capstone + teaching plan (days 43-46)				
43	Choose and plan the project	IESTI05 Part 1 / Part 2 assignments	-	One-page project plan
44	Build	-	-	Working prototype
45	Build, test and measure	-	-	Measured results (honest ones)
46	Present, write up, and plan how to teach it	-	-	Report + teaching plan
Phase 6 - Optional: Arduino UNO Q (days 47-56)				
47	UNO Q: foundations and setup	IESTI05 UNO Q Ch0-Ch1	-	Board working
50	SLMs on the UNO Q; benchmarking; Python	IESTI05 UNO Q Ch2 (C19-C21)	-	Model running
53	Multimodal and agentic AI	IESTI05 UNO Q Ch3, Ch5	-	Agent running
55	Physical AI: Bridge between the two processors	IESTI05 UNO Q Ch4	-	Language model driving hardware

NOTE Checkpoints (what you can show a student)

- **Day 5:** a Pi you can log into from either laptop, with Python and Jupyter ready.
- **Day 13:** a CNN you trained yourself. **Day 18:** your own image classifier running live on the Pi.
- **Day 26:** YOLO counting objects. **Day 32:** a question-answering system over your own documents (RAG).
- **Day 38:** a small language model switching real hardware. **Day 46:** your capstone and the plan to teach it.